PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

GND VIN_BUCK +3V3 VDRV

PWR_FLAG: these nets are fed through passive parts (diodes, R4, L1, J1), so ERC needs a flag

POWER INPUT, FUSE, LOAD-DUMP TVS, REVERSE-POLARITY PROTECTION

J1 BATTERY 12V

VBAT_IN F1 15A VBAT_F

D1 SMCJ20CA

Q1 IPD90P04P4L-04 (QG_REV, R1 10k)

D2 BZT52C12

C1 470uF 35V · C2 470uF 35V · C3 10uF 50V · C4 10uF 50V

MOTOR SWITCH: GATE DRIVER + LOW-SIDE N-MOSFET + FREEWHEEL DIODE

R4 220R · D4 BZT52C12 · C5 1uF 25V · C6 100nF

U2 UCC27517A (PWM, VDRV, GATE_DRV)

R2 10R · R3 10k · R5 100k

Q2 IPD90N04S4L-04

D3 STPS20L45CG

J2 MOTOR

3.3 V BUCK (12 V or USB)

D5 B360A · D6 B360A · C7 10uF 50V · C8 10uF 50V · C9 100nF 50V · R6 100k

U1 LMR33630ADDA

C10 100nF · L1 6.8uH

R7 100k · R8 45.2k · C12 22uF 10V · C13 22uF 10V

C11 1uF

ESP32-S3 MODULE, RESET/BOOT, BATTERY SENSE, LEDS, AUX

C14 10uF · C15 100nF

U3 ESP32-S3-WROOM-1

R9 10k · C16 1uF · SW1 RESET

R10 10k · SW2 BOOT

R13 100k · R14 18k · C17 100nF · D7 GREEN

J4 AUX

R15 1k · R16 1k · D8 RED

USB-C (PROGRAMMING / DEBUG)

J3 USB-C

U4 USBLC6-2SC6

R11 5.1k · R12 5.1k

WIPER PARK SWITCH INPUT

J5 PARK

R17 4k7 · R18 100k · R19 10k · C18 470nF

Motor: J2 pin 2 = +12 V (always on), pin 1 = MOTOR- (switched to GND by Q2).
PWM from ESP32 GPIO4, 20 kHz. R5 holds the gate driver input low while the ESP32 boots.
Battery sense: VSENSE = VBAT x 18k / 118k -> 14.4 V reads 2.20 V on GPIO1.
Q1 + D2: reverse-battery protection (P-MOSFET, gate clamped to 12 V).
D1 SMCJ20CA clamps load-dump spikes; everything on the 12 V side is rated >= 35 V.
Rated for motors up to 10 A continuous (15 A fuse). Use 2 oz copper.
J5 park input: IO7 reads PARK / 11. Firmware compares it with the battery voltage (works for contact-to-GND and 31b park switches).

H1 M3 H2 M3 H3 M3 H4 M3

Generated from design.py - board and schematic share one netlist
12 V car battery in (11-14.4 V), PWM low-side switched motor out, up to 10 A

Sheet: /		
File: motor_ctrl.kicad_sch		
Title: ESP32-S3 12 V Motor PWM Controller		
Size: A3	Date: 2026-10-26	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1